RePr: Improved Training of Convolutional Filters

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Outline

1. Motivations
2. Methods
3. Ablation Study
4. Results
5. Conclusion

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Pruned Model

Observation: a well-trained convolutional neural network can easily be pruned without significant loss of performance.

Reason: this is because of the overlap of the filters' parameters.

Observation: a model with fewer filters cannot be trained from scratch to achieve the performance of a large model that has been pruned to be roughly the same size.

Reason: Why?

Small Networks Also Learn Redundant Filters

- A simple network with two convolutional layers of 32 filters each and a softmax layer at end.
- Training this model on CIFAR-10 for 100 epochs.

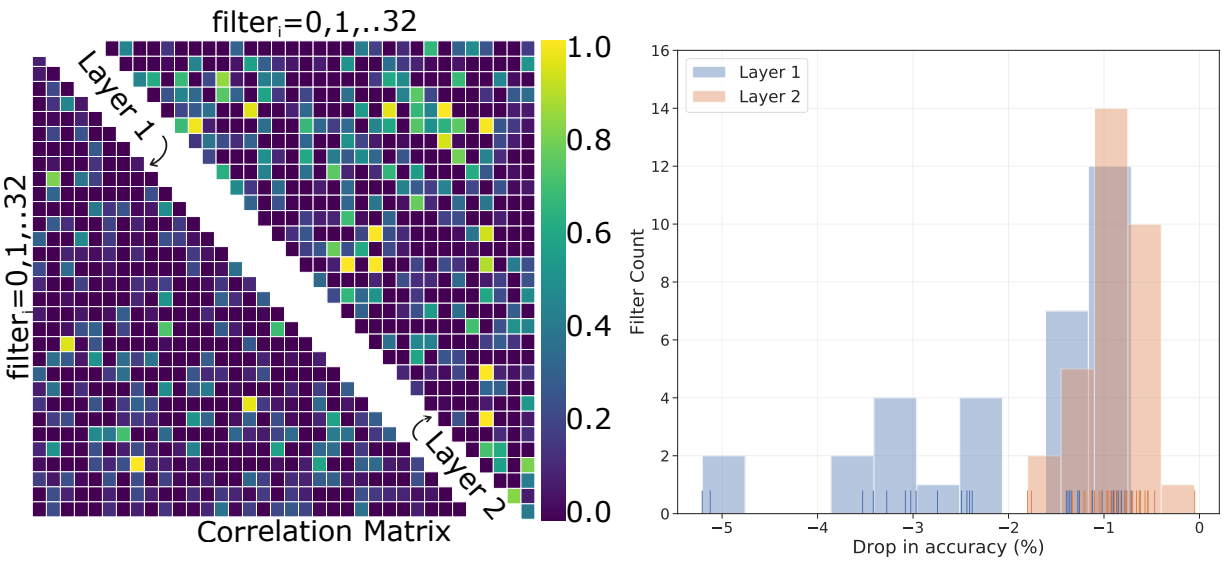

Figure 2: Left: Canonical Correlation Analysis of activations from two layers of a ConvNet trained on CIFAR-10. Right: Distribution of change in accuracy when the model is evaluated by dropping one filter at a time.

Small Networks Also Learn Redundant Filters

- The small network has clearly failed to capture the full feature space of the training data, thus any correlation is not due to the over-parametrization.

Motivation

- There is a room for improvement in the training of Convolutional Neural Networks.

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Notations

- M: a model with $\mathcal{F}$ filters spread across L layers
- $M_{\mathcal{F}}$: a complete network
- $\hat{\mathcal{F}}$: a subset of $\mathcal{F}$ filters
- $M_{\mathcal{F}-\hat{\mathcal{F}}}$: a sub-network
- $\mathcal{R}$: a metric used to rank the filters

Re-initializing and Pruning Training Scheme

Algorithm 1: RePr Training Scheme

```
for N iterations do
    for S_1 iterations do
        Train the full network: M_F
    end
    Compute the metric : R(f) ∀f ∈ F
    Let F̂ be bottom p% of F using R(f)
    for S_2 iterations do
        Train the sub-network : M_{F−F̂}
    end
    Reinitialize the filters (F̂) s.t. F̂ ⊥ F
    (and their training specific parameters
    from BatchNorm and Adam, if applicable)
end
```

Neuron Ranking Metrics

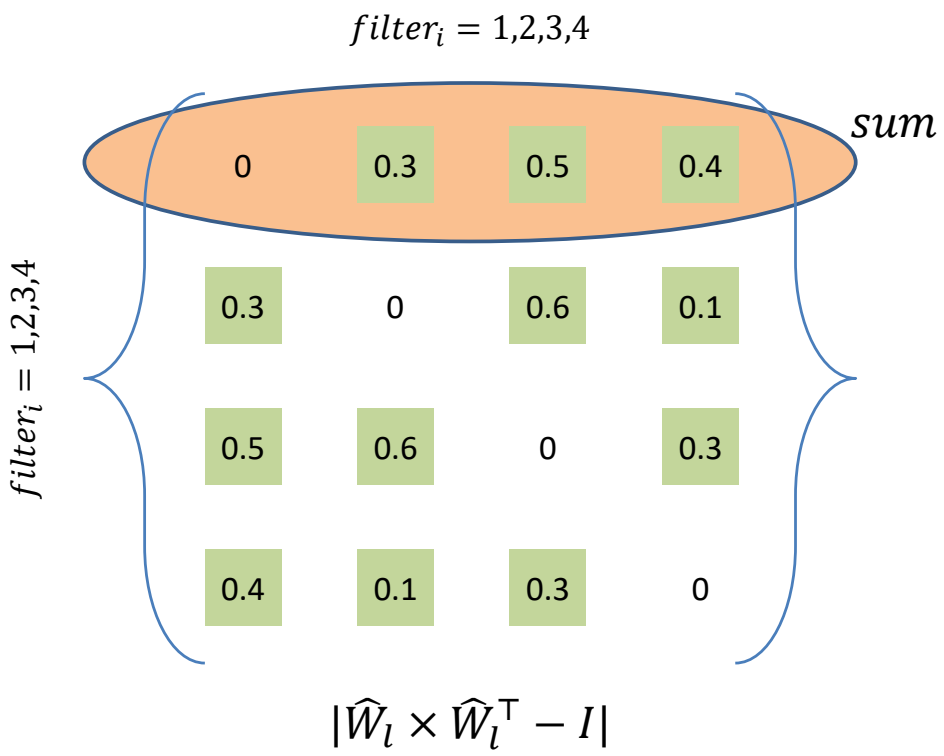
The below metrics measuring the importance of the model parameter.

- Hessian (contains second-order derivative, while computation is expensive)
- The norm of weights (yields sparse models)
- Average percentage of zeros (APoZ) in the activations
- The second term from the Taylor expansion of the loss function
- Greedy Oracle: the drop accuracy on test set by removing each filter (not computationally feasible for large networks)

Inter-Filter Orthogonality Metric

Notations: Let $f \in \mathbb{R}^{k \times k \times c}$ be a 3D tensor filter. Let $W_\ell \in \mathbb{R}^{J_\ell \times k * k * c}$ be the filter matrix of ℓ-th layer which is flatten from a 4D tensor, and J_ℓ is the number of filters in the layer ℓ.

The measure of orthogonality for filter f in layer ℓ (denoted by O_ℓ^f) is computed as shown in equations below.



$$P_\ell = |\hat{W}_\ell \times \hat{W}_\ell^\top - I|$$

$$\hat{W}_\ell = W_\ell / ||W_\ell||$$

$$O_\ell^f = \frac{\sum P_\ell[f]}{J_\ell}$$

Performance of Inter-Filter Orthogonality Metric

Computation complexity:

the norm of weights, average percentage of zeros $<$ **Ortho** $<$ Hessian

Performance of different metrics for training a 3 layer ConvNet with each layer has 32 filters on CIFAR-10.

CIFAR-10 - $C^3(32)$	
Standard	72.1
Random	73.4
Activations	74.1
APoZ [24]	74.3
Gradients [22]	74.3
Taylor [9]	74.3
Hessian [22]	74.4
Weights [12]	74.6
Oracle	76.0
Ortho	**76.4**

Dynamics and Impacts of RePr

Consider a n layer vanilla ConvNet $C^n(X)$, with X filters each layer:

$$\text{Img} \mapsto [\text{CONV}(X) \rightarrow \text{RELU}]^n \mapsto \text{FC} \mapsto \text{Softmax} \tag{1}$$

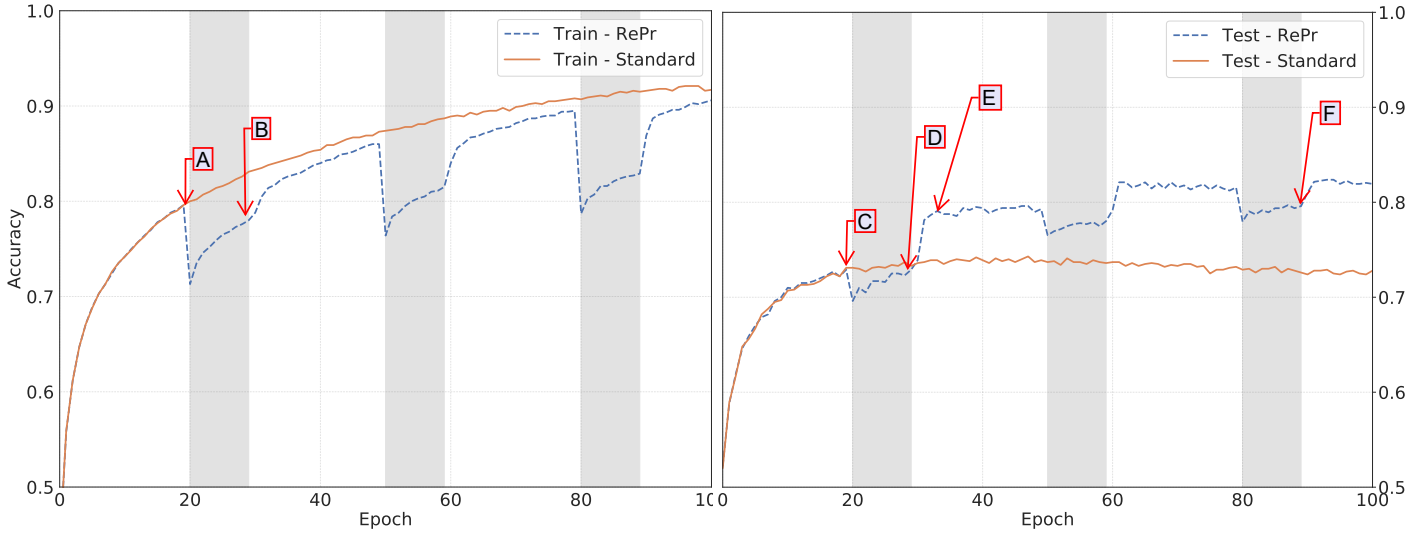


Figure: 1) The shaded regions denote periods when only part of the network is trained. 2) Left: Training Accuracy, Right: Test Accuracy.

Dynamics and Impacts of RePr

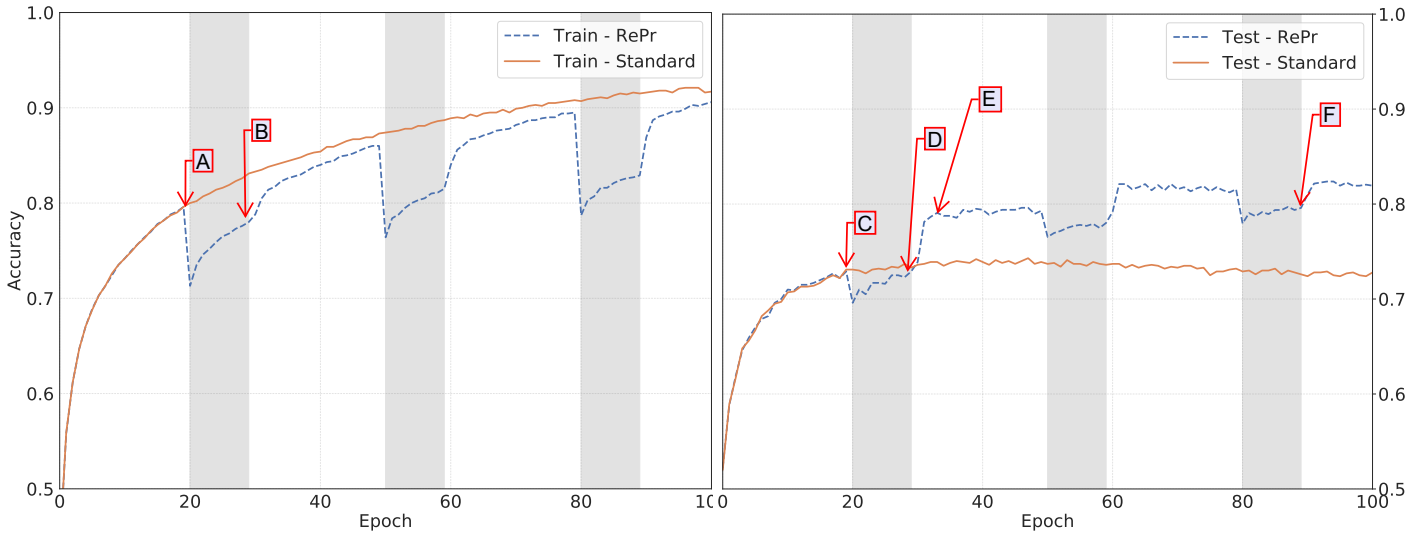


Figure: 1) The shaded regions denote periods when only part of the network is trained. 2) Left: Training Accuracy, Right: Test Accuracy.

- The training accuracy of RePr model is lower, which signifies some form of regularization on the model.
- The test accuracy at [D] is the same as [C] but at this point, the model only has 70% filters.
- The test accuracy at [E] is significantly higher than [C]. Both point [C] and [E] are same capacity networks.

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Comparsion of Pruning Criteria

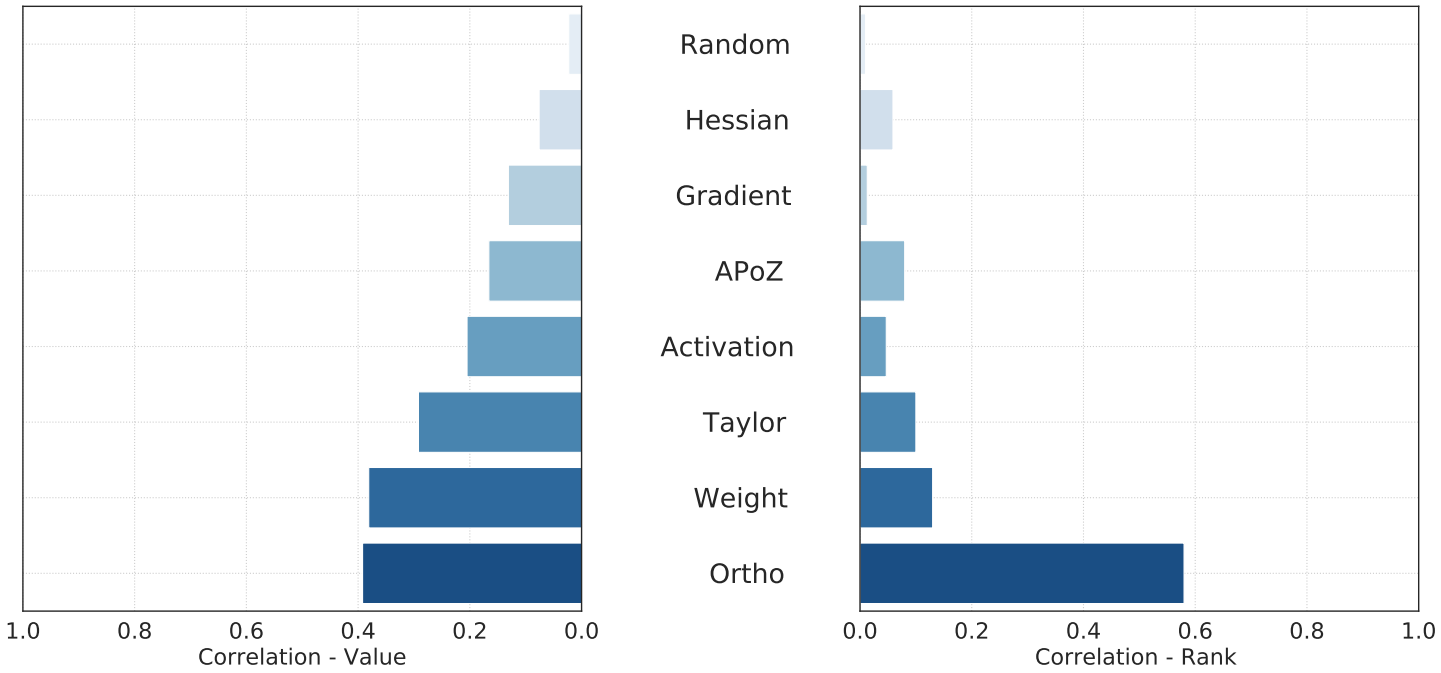
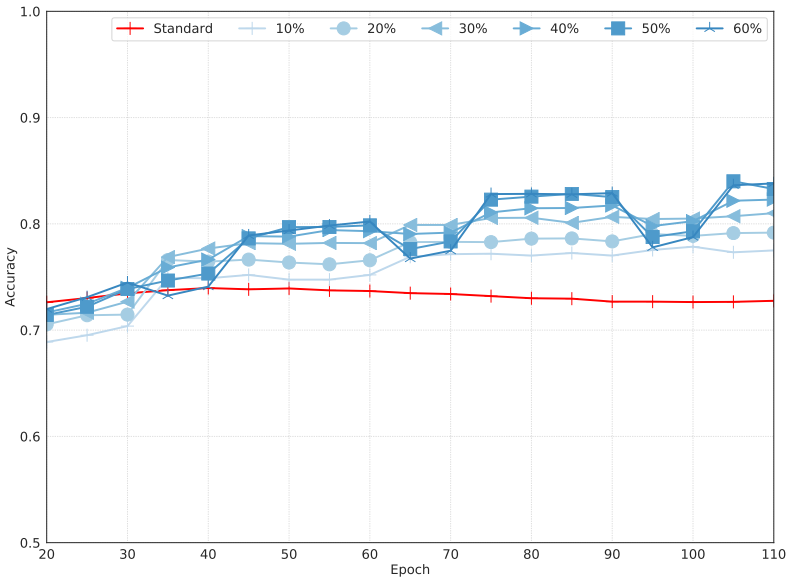


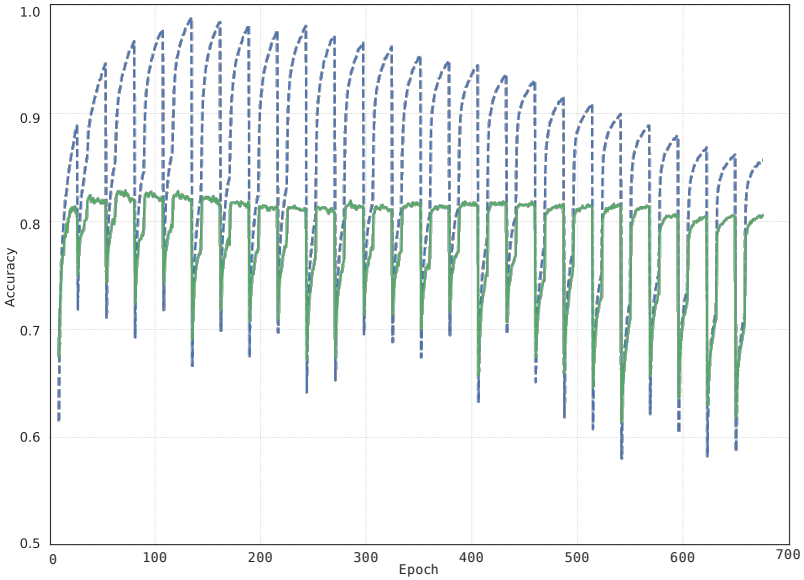
Figure: **Left:** Pearson correlation coefficient of various metric values with the accuracy values from greedy oracle. **Right:** Pearson correlation coefficient of filter ranks using various metric with rank from greedy oracle.

Percentage of Filters Pruned



- In general, the higher the pruned percentage, the better the performance.
- When beyond 30%, the performances are not significant.
- Up to 50%, the model seems to recover from the droping of filters and the training is not stable.

Number of RePr Iterations



- Training and Test accuracy.
- Two to three iterations is sufficient to achieve the maximum benefit.
- With the increase of iterations, the generalization gap between train and test accuracy is reduced significantly.

Performance of Orthogonal Loss

Adding orthogonality of filters as regularization term as a part of the optimization loss directly:

$$\mathcal{L} = \text{Cross entropy} + \lambda * |\hat{W}_\ell \times \hat{W}_\ell^\top - I| \tag{2}$$

$C^3(32)$	**Std**	**Std+OL**	**RePr**	**RePr+OL**
CIFAR-10	72.1	72.8	76.4	76.7
CIFAR-100	47.2	48.3	58.2	58.6

Table 1: Comparison of addition of Orthogonality loss to Standard Training and RePr

Equation (2) does not significantly impact the performance of the model.

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Comparisons with Other Training Schemes

ResNet-20 on CIFAR-10					
Baseline		Various Training Schemes			
Original [1]	**Our Impl**	**DSD [18]**	**BAN [20]**	**RePr Weights**	**RePr Ortho**
8.7	8.4	7.8	8.2	7.7	6.9

Table 3: Comparison of test error from using various techniques.

- **Remark:** RePr training acts on entire filters while DSD acts on individual weights.

Comparisons with Modern Architectures

			CIFAR-10		CIFAR-100	
Layers	Params (×000)	Inf. Time (relative)	Std	RePr	Std	RePr
Vanilla CNN [32 filters / layer]						
3	20	1.0	27.9	23.6	52.8	41.8
8	66	1.7	26.8	19.5	50.9	36.8
13	113	2.5	26.6	20.6	51.0	37.9
18	159	3.3	28.2	22.5	51.9	39.5
DenseNet [k=12]						
5	1.7	0.9	39.4	36.2	43.5	40.9
40	1016	8.0	6.8	6.2	26.4	25.2
100	6968	43.9	5.3	5.6	22.2	22.1
ResNet						
20	269	1.7	8.4	6.9	32.6	31.1
32	464	2.2	7.4	6.1	31.4	30.1
110	1727	7.1	6.3	5.4	27.5	26.4
182	2894	11.7	5.6	5.1	26.0	25.3

Table 4: Comparison of test error on Cifar-10 & Cifar-100 of various ConvNets using Standard training vs RePr Training. Inf. Time shows the inference times for a single pass. All time measurements are relative to Vanilla CNN with three layers. Parameter count does not include the last fully-connected layer.

- Vanilla CNNs are significantly more inefficiency in the feature extraction, thus the improvements from RePr is larger.
- RePr also improves the performance of modern architectures (DenseNet, ResNet, etc.)
- For the same inference time (3 layer Vanilla CNN and 5 layer DenseNet), DenseNet has high error rates.

Comparisons of Vanilla CNNs with Different Depths

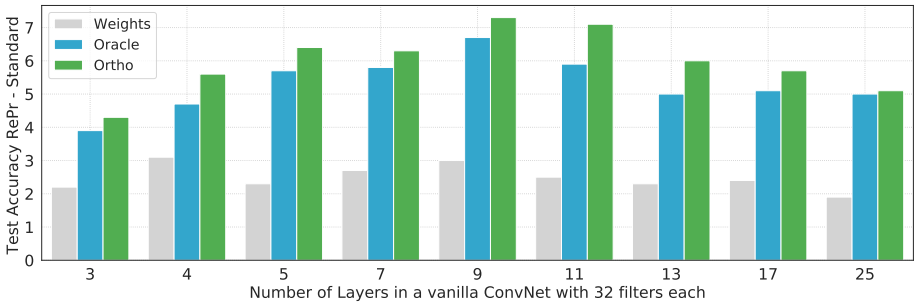

Figure 9: Accuracy improvement using RePr over standard training on Vanilla ConvNets across many layered networks $[C^n(32)]$

			CIFAR-10		CIFAR-100	
Layers	Params ($\times$000)	Inf. Time (relative)	Std	RePr	Std	RePr
Vanilla CNN [32 filters / layer]						
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18	159	3.3	28.2	22.5	51.9	39.5

- Vanilla CNNs starts to overfit the data, as most filters converge to similar representation, and the improvements start to decrease.
- RePr forces the model parameters to be different which reduces the overfitting. The evident is that the larger test error of 18-layer vanilla CNN with CIFAR-10 compared to a 3-layer CNN and the lower test error.

Performance of RePr on ImageNet

ImageNet					
Model	Standard Training	RePr Training			Relative Change
		N=1	N=2	N=3	
ResNet-18	30.41	28.68	27.87	27.31	-11.35
ResNet-34	27.50	26.49	26.06	25.80	-6.59
ResNet-50	23.67	22.79	22.51	22.37	-5.81
ResNet-101	22.40	21.70	21.51	21.40	-4.67
ResNet-152	21.51	20.99	20.79	20.71	-3.86
VGG-16	31.30	27.76	26.45	25.50	-22.75
Inception v1	31.11	29.41	28.47	28.01	-11.07
Inception v2	27.60	27.15	26.95	26.80	-2.99

Table 5: Comparison of test error (Top-1) on ImageNet with different models at various stages of RePr. N=1, N=2, N=3 are results after each round of RePr.

- Each iteration leads to improved performance with diminishing returns.
- Improvement is more distinct in architectures which do not have skip connections (like Inception v1 and VGG).

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Conclusion

- The motivation is clear. Based on that, the authors proposed a training scheme to reduce the redundancy of filters.
- The proposed filter importance metric can achieve the best performance under the RePr training scheme.
- The experiments in this paper fully showed the efficiency and robustness of the proposed method.
- For many problems and observations in the experiments, the authors gave their own reasonable explanation.
- The authors tried to comprehend their proposed method from varying aspects.

Thank you for your attention!